

OFFICE NETWORK BUILD

OSPF

Backstory:

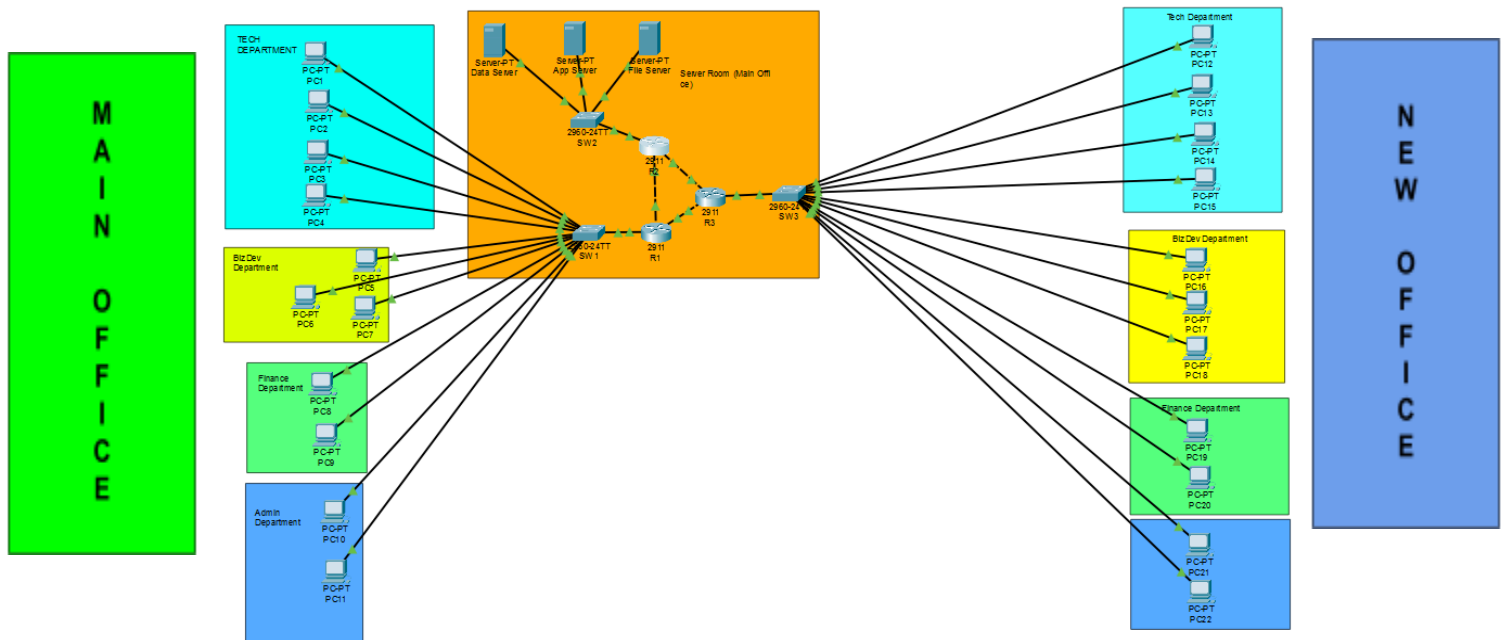
After successfully establishing connectivity between the company's different departments and server room using static and default routing, the company experienced further growth. The organization was able to acquire a new office space located in the same building, directly beside the existing main office. The newly acquired space was the office that had recently been vacated, giving the company an opportunity to expand without having to relocate to another building.

The existing server room remains inside the main office, where the company's servers and core network infrastructure are housed. To provide network connectivity to the newly acquired office, network cabling was routed from the new office toward the server room through the building's movable ceiling panels. This allowed the new office to share access to the company's existing network infrastructure without requiring the construction of a separate server room.

However, the expansion also introduced a new networking challenge. The company now has additional networks and another office that must communicate with the existing infrastructure. As the network continues to grow, relying solely on manually configured static routes becomes increasingly difficult to maintain. Every time another network is added, administrators would need to manually configure routes on the appropriate routers.

Because the company expects the network to continue expanding, a dynamic routing protocol is now more appropriate. For this phase of the project, Open Shortest Path First (OSPF) will be introduced to allow the routers to dynamically discover and exchange information about the networks available throughout the company's infrastructure.

Current Topology:



Why OSPF?

As the company expands and adds the new office, the network now contains more routers and additional networks that need to communicate with one another. While static routes worked well for the original, smaller topology, manually configuring routes on every router becomes increasingly difficult to manage as the network grows. OSPF (Open Shortest Path First) provides a more scalable solution by allowing routers to dynamically exchange information about the networks they can reach and automatically determine the best available path. This is particularly useful for the company's expanded network because changes such as adding new networks or links can be propagated throughout the routing domain without requiring every route to be manually configured. OSPF therefore makes the network more scalable, easier to maintain, and better prepared for future expansion.

The image below shows the whole IP Address table of all the equipments in the two offices:

Server Room					
Device	IP	Subnet	Default Gateway	VLAN	Interface
R1	192.168.100.1	255.255.255.248 - /29	N/A	VLAN 10	G0/0.10
R1	192.168.100.9	255.255.255.248 - /29	N/A	VLAN 20	G0/0.20
R1	192.168.100.17	255.255.255.248 - /29	N/A	VLAN 30	G0/0.30
R1	192.168.100.25	255.255.255.248 - /29	N/A	VLAN 40	G0/0.40
R1 (R1-R2 Transit Link)	192.168.254.1	255.255.255.252 - /30	N/A	N/A	G0/1
R1 (R1-R3 Transit Link)	192.168.254.10		N/A	N/A	G0/2
R2	192.168.101.1	255.255.255.0 - /24	N/A	N/A	G0/0
R2 (Transit Link)	192.168.254.2	255.255.255.252 - /30	N/A	N/A	G0/1
R2 (Transit Link)	192.168.254.5	255.255.255.252 - /30	N/A	N/A	G0/2
R3	192.168.102.1	255.255.255.248 - /29	N/A	VLAN 10	G0/0.10
R3	192.168.102.9	255.255.255.248 - /29	N/A	VLAN 20	G0/0.20
R3	192.168.102.17	255.255.255.248 - /29	N/A	VLAN 30	G0/0.30
R3	192.168.102.25	255.255.255.248 - /29	N/A	VLAN 40	G0/0.40
R3 (R3-R2 Transit Link)	192.168.254.6	255.255.255.252 - /30	N/A	N/A	G0/1
R3 (R3-R1 Transit Link)	192.168.254.9	255.255.255.252 - /30	N/A	N/A	G0/2
Data Server	192.168.101.10	255.255.255.0 - /24	N/A	N/A	Fa0
Application Server	192.168.101.20	255.255.255.0 - /24	N/A	N/A	Fa0
File Server	192.168.101.30	255.255.255.0 - /24	N/A	N/A	Fa0
Main Office					
Device	IP	Subnet	Default Gateway	VLAN	Interface
PC1	192.168.100.2	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC2	192.168.100.3	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC3	192.168.100.4	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC4	192.168.100.5	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC5	192.168.100.10	255.255.255.248 - /29	192.168.100.9	VLAN 20	Fa0
PC6	192.168.100.11	255.255.255.248 - /29	192.168.100.9	VLAN 20	Fa0
PC7	192.168.100.12	255.255.255.248 - /29	192.168.100.9	VLAN 20	Fa0
PC8	192.168.100.18	255.255.255.248 - /29	192.168.100.17	VLAN 30	Fa0
PC9	192.168.100.19	255.255.255.248 - /29	192.168.100.17	VLAN 30	Fa0
PC10	192.168.100.26	255.255.255.248 - /29	192.168.100.25	VLAN 40	Fa0
PC11	192.168.100.27	255.255.255.248 - /29	192.168.100.25	VLAN 40	Fa0
2nd Office					
Device	IP	Subnet	Default Gateway	VLAN	Interface
PC12	192.168.103.2	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC13	192.168.103.3	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC14	192.168.103.4	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC15	192.168.103.5	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC16	192.168.103.10	255.255.255.248 - /29	192.168.100.9	VLAN 20	Fa0
PC17	192.168.103.11	255.255.255.248 - /29	192.168.100.9	VLAN 20	Fa0
PC18	192.168.103.12	255.255.255.248 - /29	192.168.100.9	VLAN 20	Fa0
PC19	192.168.103.18	255.255.255.248 - /29	192.168.100.17	VLAN 30	Fa0
PC20	192.168.103.19	255.255.255.248 - /29	192.168.100.17	VLAN 30	Fa0
PC21	192.168.103.26	255.255.255.248 - /29	192.168.100.25	VLAN 40	Fa0

Network Addressing

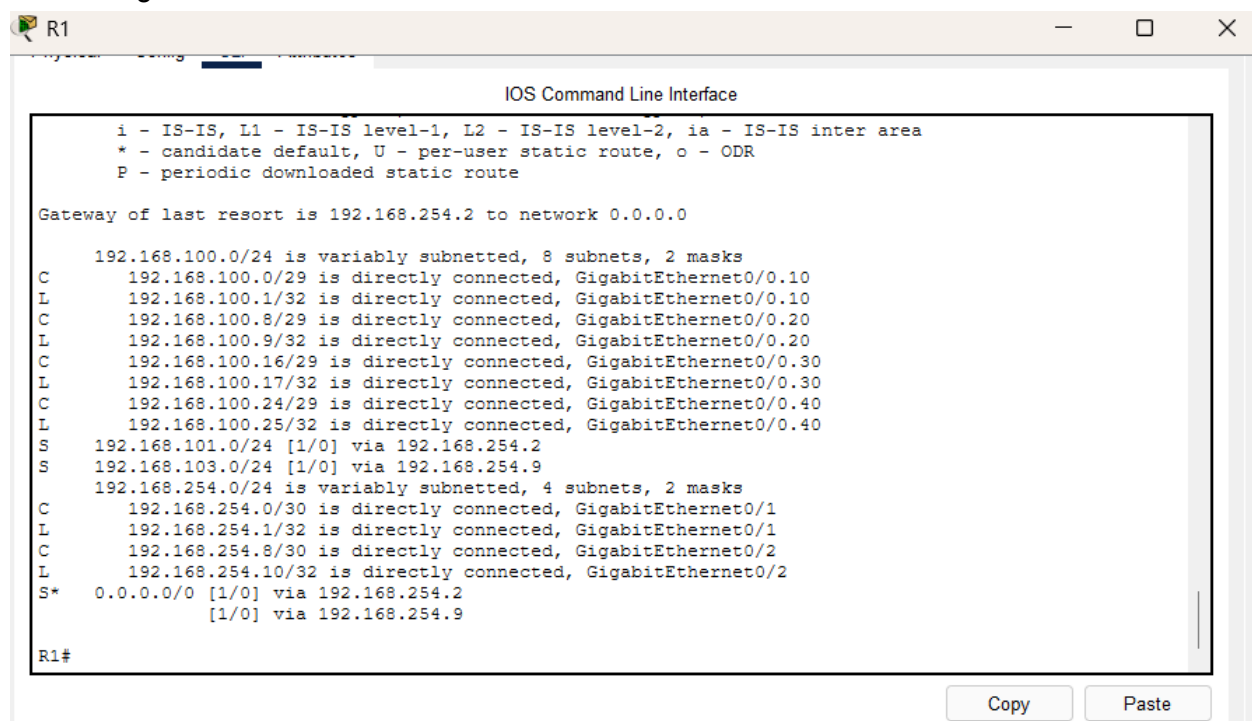
To support the company's expansion, the network was organized into three primary network segments. The Main Office PCs are assigned to the 192.168.100.0 network, where the existing departmental VLANs for Tech, Business Development, Finance, and Admin are located. The company's servers, including the Data Server, Application Server, and File Server, are placed on the separate 192.168.101.0 network within the server room. Meanwhile, the newly acquired 2nd Office uses the 192.168.102.0 network for its PCs and departmental VLANs. Separating these locations into distinct networks provides a clear structure for routing between the Main Office,

Server Room, and 2nd Office. With these networks now distributed across the company's routers, OSPF will be used to dynamically exchange their routes and maintain connectivity between the different network segments as the infrastructure continues to grow.

Current Configured Routes:

Before proceeding with OSPF, the company's network is currently relying on static routes and default routes to provide connectivity between the Main Office, Server Room, and newly acquired 2nd Office. While these routing methods have successfully established communication between the different networks, they require routes to be manually configured and maintained on the routers. To demonstrate the advantages of dynamic routing, we will remove the existing static and default routes and test the network before implementing OSPF. This will provide a clear baseline and allow us to verify that connectivity between the different networks can later be restored through OSPF dynamic route learning rather than manually configured routes.

R1 Routing Table:

A screenshot of a Cisco IOS Command Line Interface window titled 'R1'. The window shows the routing table for R1. The output includes a legend for route types (i, *, P), a gateway of last resort, and a list of routes with their sources and next hops. The routes are categorized by their source: 192.168.100.0/24 (8 subnets), 192.168.254.0/24 (4 subnets), and a default route (0.0.0.0/0).

```
R1# show ip route
IOS Command Line Interface

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 192.168.254.2 to network 0.0.0.0

192.168.100.0/24 is variably subnetted, 8 subnets, 2 masks
C    192.168.100.0/29 is directly connected, GigabitEthernet0/0.10
L    192.168.100.1/32 is directly connected, GigabitEthernet0/0.10
C    192.168.100.8/29 is directly connected, GigabitEthernet0/0.20
L    192.168.100.9/32 is directly connected, GigabitEthernet0/0.20
C    192.168.100.16/29 is directly connected, GigabitEthernet0/0.30
L    192.168.100.17/32 is directly connected, GigabitEthernet0/0.30
C    192.168.100.24/29 is directly connected, GigabitEthernet0/0.40
L    192.168.100.25/32 is directly connected, GigabitEthernet0/0.40
S    192.168.101.0/24 [1/0] via 192.168.254.2
S    192.168.103.0/24 [1/0] via 192.168.254.9
192.168.254.0/24 is variably subnetted, 4 subnets, 2 masks
C    192.168.254.0/30 is directly connected, GigabitEthernet0/1
L    192.168.254.1/32 is directly connected, GigabitEthernet0/1
C    192.168.254.8/30 is directly connected, GigabitEthernet0/2
L    192.168.254.10/32 is directly connected, GigabitEthernet0/2
S*   0.0.0.0/0 [1/0] via 192.168.254.2
      [1/0] via 192.168.254.9

R1#
```

R2 Routing Table:

```
R2
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.254.1 to network 0.0.0.0

S    192.168.100.0/24 [1/0] via 192.168.254.1
    192.168.101.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.101.0/24 is directly connected, GigabitEthernet0/0
L    192.168.101.1/32 is directly connected, GigabitEthernet0/0
S    192.168.102.0/24 [1/0] via 192.168.254.6
    192.168.254.0/24 is variably subnetted, 4 subnets, 2 masks
C    192.168.254.0/30 is directly connected, GigabitEthernet0/1
L    192.168.254.2/32 is directly connected, GigabitEthernet0/1
C    192.168.254.4/30 is directly connected, GigabitEthernet0/2
L    192.168.254.5/32 is directly connected, GigabitEthernet0/2
S*   0.0.0.0/0 [1/0] via 192.168.254.1
      [1/0] via 192.168.254.6

R2#
```

☐ Top

R3 Routing Table:

```
R3
Gateway of last resort is 192.168.254.5 to network 0.0.0.0

S    192.168.100.0/24 [1/0] via 192.168.254.10
      [1/0] via 192.168.102.10
S    192.168.101.0/24 [1/0] via 192.168.254.5
      [1/0] via 192.168.102.5
    192.168.102.0/24 is variably subnetted, 8 subnets, 2 masks
C    192.168.102.0/29 is directly connected, GigabitEthernet0/0.10
L    192.168.102.1/32 is directly connected, GigabitEthernet0/0.10
C    192.168.102.8/29 is directly connected, GigabitEthernet0/0.20
L    192.168.102.9/32 is directly connected, GigabitEthernet0/0.20
C    192.168.102.16/29 is directly connected, GigabitEthernet0/0.30
L    192.168.102.17/32 is directly connected, GigabitEthernet0/0.30
C    192.168.102.24/29 is directly connected, GigabitEthernet0/0.40
L    192.168.102.25/32 is directly connected, GigabitEthernet0/0.40
    192.168.254.0/24 is variably subnetted, 4 subnets, 2 masks
C    192.168.254.4/30 is directly connected, GigabitEthernet0/1
L    192.168.254.6/32 is directly connected, GigabitEthernet0/1
C    192.168.254.8/30 is directly connected, GigabitEthernet0/2
L    192.168.254.9/32 is directly connected, GigabitEthernet0/2
S*   0.0.0.0/0 [1/0] via 192.168.254.5
      [1/0] via 192.168.254.10

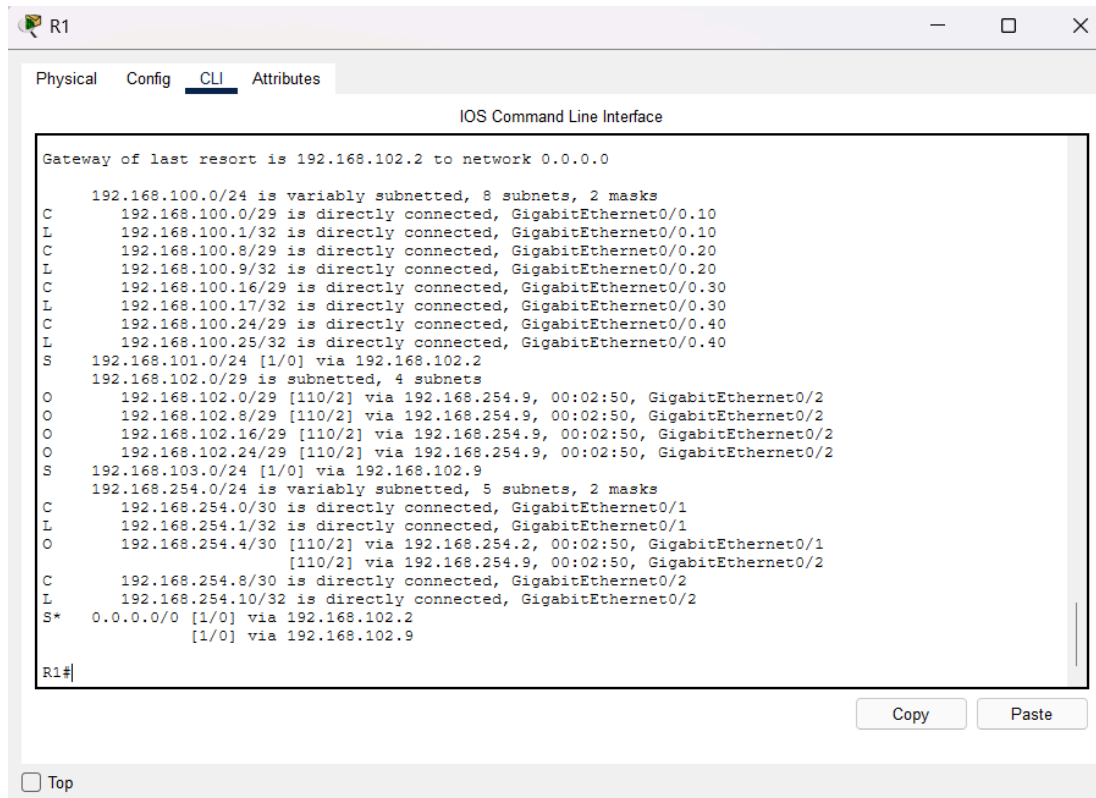
R3#
```

☐ Top

Configuring and Testing OSPF

With the existing static and default routes removed, OSPF was configured on each router using OSPF Process ID 1 and Area 0. The directly connected networks, including the Main Office, Server Room, 2nd Office, and router-to-router transit networks, were added to the OSPF process using their corresponding wildcard masks. After configuration, the OSPF neighbor relationships and routing tables were checked to verify that the routers were successfully exchanging network information. Connectivity tests were then performed between the different network segments to confirm that routes were being learned dynamically through OSPF rather than through manually configured static or default routes.

R1 Routing Table:



The screenshot shows the R1 routing table in Cisco Packet Tracer. The window title is 'R1' and it has tabs for 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is active, showing the 'IOS Command Line Interface'. The routing table output is as follows:

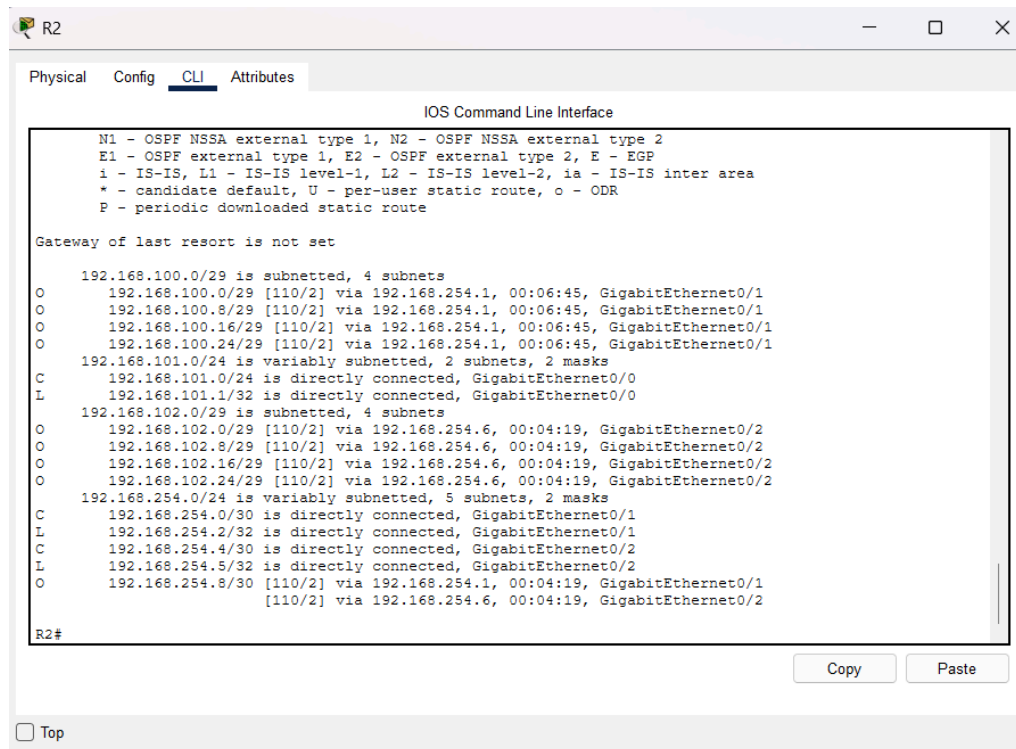
```
Gateway of last resort is 192.168.102.2 to network 0.0.0.0

  192.168.100.0/24 is variably subnetted, 8 subnets, 2 masks
C    192.168.100.0/29 is directly connected, GigabitEthernet0/0.10
L    192.168.100.1/32 is directly connected, GigabitEthernet0/0.10
C    192.168.100.8/29 is directly connected, GigabitEthernet0/0.20
L    192.168.100.9/32 is directly connected, GigabitEthernet0/0.20
C    192.168.100.16/29 is directly connected, GigabitEthernet0/0.30
L    192.168.100.17/32 is directly connected, GigabitEthernet0/0.30
C    192.168.100.24/29 is directly connected, GigabitEthernet0/0.40
L    192.168.100.25/32 is directly connected, GigabitEthernet0/0.40
S    192.168.101.0/24 [1/0] via 192.168.102.2
  192.168.102.0/29 is subnetted, 4 subnets
O    192.168.102.0/29 [110/2] via 192.168.254.9, 00:02:50, GigabitEthernet0/2
O    192.168.102.8/29 [110/2] via 192.168.254.9, 00:02:50, GigabitEthernet0/2
O    192.168.102.16/29 [110/2] via 192.168.254.9, 00:02:50, GigabitEthernet0/2
O    192.168.102.24/29 [110/2] via 192.168.254.9, 00:02:50, GigabitEthernet0/2
S    192.168.103.0/24 [1/0] via 192.168.102.9
  192.168.254.0/24 is variably subnetted, 5 subnets, 2 masks
C    192.168.254.0/30 is directly connected, GigabitEthernet0/1
L    192.168.254.1/32 is directly connected, GigabitEthernet0/1
O    192.168.254.4/30 [110/2] via 192.168.254.2, 00:02:50, GigabitEthernet0/1
      [110/2] via 192.168.254.9, 00:02:50, GigabitEthernet0/2
C    192.168.254.8/30 is directly connected, GigabitEthernet0/2
L    192.168.254.10/32 is directly connected, GigabitEthernet0/2
S*   0.0.0.0/0 [1/0] via 192.168.102.2
      [1/0] via 192.168.102.9

R1#
```

At the bottom of the CLI window, there are 'Copy' and 'Paste' buttons. Below the CLI window, there is a 'Top' button with a checkbox.

R2 Routing Table:



The screenshot shows the R2 router's routing table in the Packet Tracer interface. The window is titled 'R2' and has tabs for 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is active, displaying the 'IOS Command Line Interface'. The routing table is shown as a list of routes with their status, next hop, and interface. The routes are as follows:

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

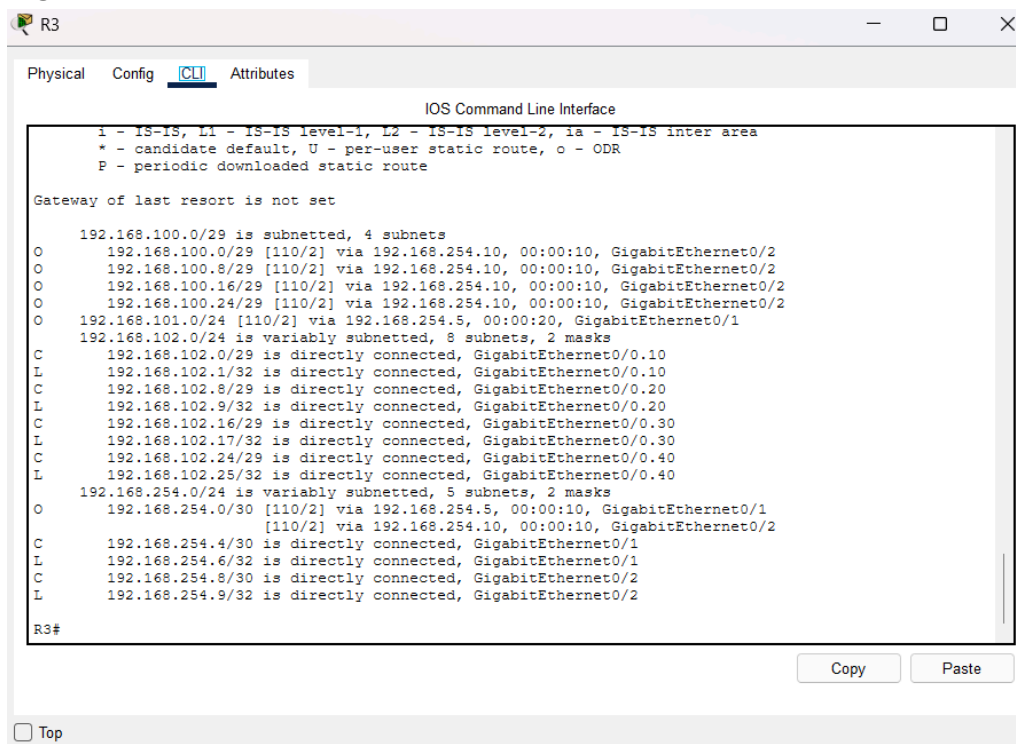
Gateway of last resort is not set

192.168.100.0/29 is subnetted, 4 subnets
O 192.168.100.0/29 [110/2] via 192.168.254.1, 00:06:45, GigabitEthernet0/1
O 192.168.100.8/29 [110/2] via 192.168.254.1, 00:06:45, GigabitEthernet0/1
O 192.168.100.16/29 [110/2] via 192.168.254.1, 00:06:45, GigabitEthernet0/1
O 192.168.100.24/29 [110/2] via 192.168.254.1, 00:06:45, GigabitEthernet0/1
192.168.101.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.101.0/24 is directly connected, GigabitEthernet0/0
L 192.168.101.1/32 is directly connected, GigabitEthernet0/0
192.168.102.0/29 is subnetted, 4 subnets
O 192.168.102.0/29 [110/2] via 192.168.254.6, 00:04:19, GigabitEthernet0/2
O 192.168.102.8/29 [110/2] via 192.168.254.6, 00:04:19, GigabitEthernet0/2
O 192.168.102.16/29 [110/2] via 192.168.254.6, 00:04:19, GigabitEthernet0/2
O 192.168.102.24/29 [110/2] via 192.168.254.6, 00:04:19, GigabitEthernet0/2
192.168.254.0/24 is variably subnetted, 5 subnets, 2 masks
C 192.168.254.0/30 is directly connected, GigabitEthernet0/1
L 192.168.254.2/32 is directly connected, GigabitEthernet0/1
C 192.168.254.4/30 is directly connected, GigabitEthernet0/2
L 192.168.254.5/32 is directly connected, GigabitEthernet0/2
O 192.168.254.8/30 [110/2] via 192.168.254.1, 00:04:19, GigabitEthernet0/1
[110/2] via 192.168.254.6, 00:04:19, GigabitEthernet0/2

R2#
```

At the bottom of the CLI window, there are 'Copy' and 'Paste' buttons. Below the CLI window, there is a 'Top' button.

R3 Routing Table:



The screenshot shows the R3 router's routing table in the Packet Tracer interface. The window is titled 'R3' and has tabs for 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is active, displaying the 'IOS Command Line Interface'. The routing table is shown as a list of routes with their status, next hop, and interface. The routes are as follows:

```
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

192.168.100.0/29 is subnetted, 4 subnets
O 192.168.100.0/29 [110/2] via 192.168.254.10, 00:00:10, GigabitEthernet0/2
O 192.168.100.8/29 [110/2] via 192.168.254.10, 00:00:10, GigabitEthernet0/2
O 192.168.100.16/29 [110/2] via 192.168.254.10, 00:00:10, GigabitEthernet0/2
O 192.168.100.24/29 [110/2] via 192.168.254.10, 00:00:10, GigabitEthernet0/2
O 192.168.101.0/24 [110/2] via 192.168.254.5, 00:00:20, GigabitEthernet0/1
192.168.102.0/24 is variably subnetted, 8 subnets, 2 masks
C 192.168.102.0/29 is directly connected, GigabitEthernet0/0.10
L 192.168.102.1/32 is directly connected, GigabitEthernet0/0.10
C 192.168.102.8/29 is directly connected, GigabitEthernet0/0.20
L 192.168.102.9/32 is directly connected, GigabitEthernet0/0.20
C 192.168.102.16/29 is directly connected, GigabitEthernet0/0.30
L 192.168.102.17/32 is directly connected, GigabitEthernet0/0.30
C 192.168.102.24/29 is directly connected, GigabitEthernet0/0.40
L 192.168.102.25/32 is directly connected, GigabitEthernet0/0.40
192.168.254.0/24 is variably subnetted, 5 subnets, 2 masks
O 192.168.254.0/30 [110/2] via 192.168.254.5, 00:00:10, GigabitEthernet0/1
[110/2] via 192.168.254.10, 00:00:10, GigabitEthernet0/2
C 192.168.254.4/30 is directly connected, GigabitEthernet0/1
L 192.168.254.6/32 is directly connected, GigabitEthernet0/1
C 192.168.254.8/30 is directly connected, GigabitEthernet0/2
L 192.168.254.9/32 is directly connected, GigabitEthernet0/2

R3#
```

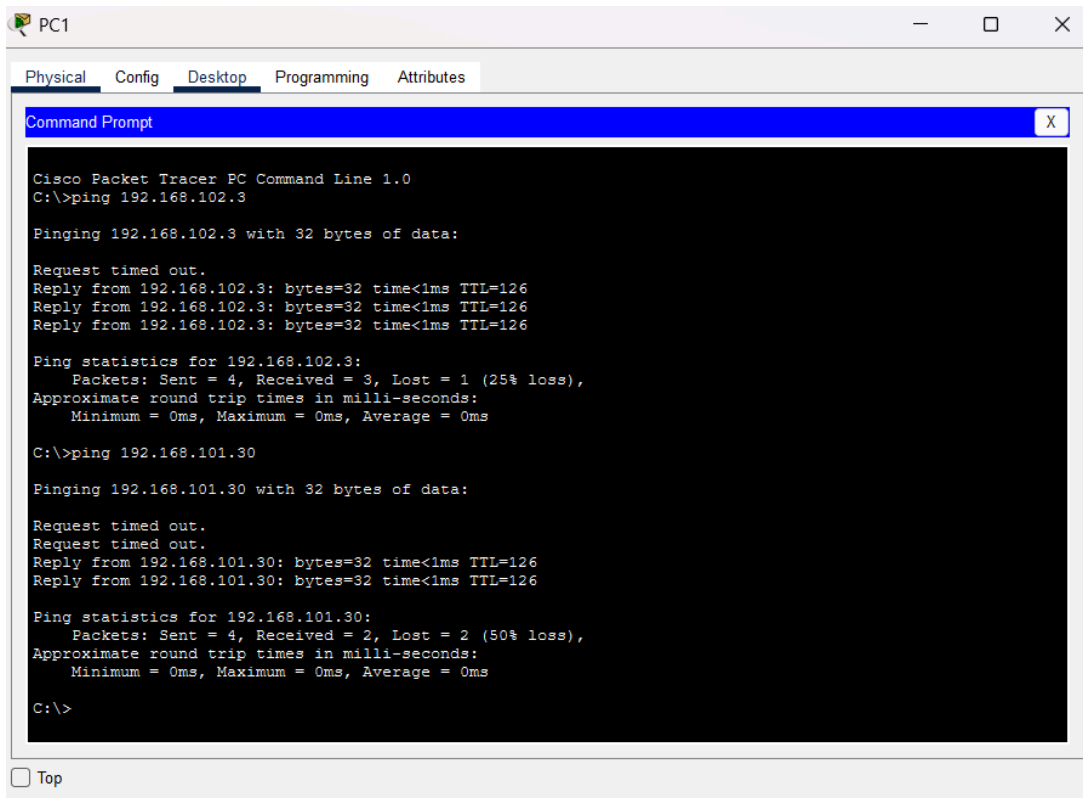
At the bottom of the CLI window, there are 'Copy' and 'Paste' buttons. Below the CLI window, there is a 'Top' button.

Once all routers were configured with OSPF and all directly connected networks were associated with the appropriate router, the routes were automatically learned and appeared in each router's routing table. What was particularly interesting was that OSPF was able to recognize not only the major networks, such as 192.168.100.0 and 192.168.102.0, but also their individual subnets. This demonstrated how OSPF dynamically identifies and advertises the specific networks available throughout the topology, eliminating the need to manually configure each route on every router.

The Path:

The routing table shows several routes marked with "O," which indicates that the route was learned dynamically through OSPF. For example, the entry O 192.168.102.0/29 [110/2] via 192.168.254.9, GigabitEthernet0/2 tells R1 that the 192.168.102.0/29 network can be reached through the next-hop address 192.168.254.9 using GigabitEthernet0/2. When R1 receives a packet destined for a device within this subnet, it checks the routing table, identifies the OSPF-learned route, and forwards the packet through GigabitEthernet0/2 toward the next-hop router. This demonstrates how OSPF automatically provides R1 with the path to remote networks without requiring a manually configured static route.

PC1 (Main Office - Tech Dept.) pinging random PC and server in New Office:



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.102.3

Pinging 192.168.102.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.102.3: bytes=32 time<1ms TTL=126
Reply from 192.168.102.3: bytes=32 time<1ms TTL=126
Reply from 192.168.102.3: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.102.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.101.30

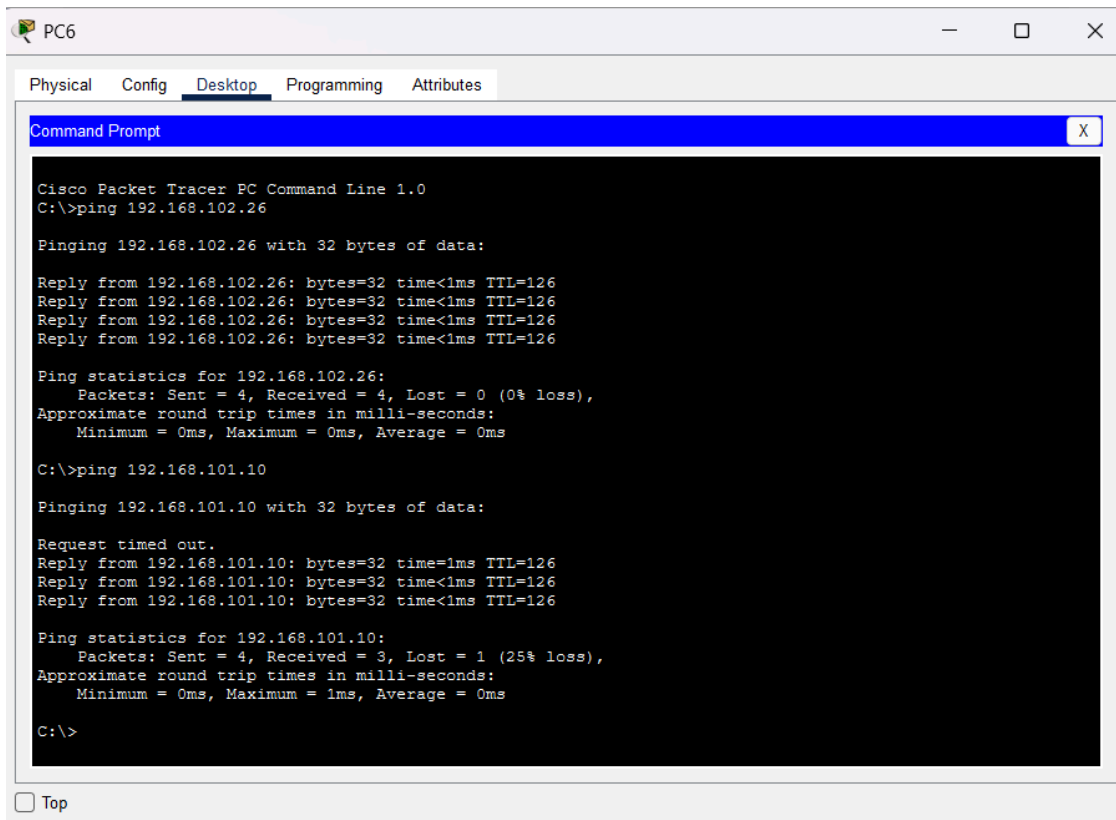
Pinging 192.168.101.30 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.30:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PC6 (Main Office - BizDev Dept.) pinging random PC and server in New Office:



The screenshot shows a Cisco Packet Tracer interface with a window titled 'PC6'. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of two ping commands. The first command, 'ping 192.168.102.26', results in four successful replies with 0% loss. The second command, 'ping 192.168.101.10', results in three successful replies and one request timed out, resulting in a 25% loss. The window has a 'Top' button at the bottom left.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.102.26

Pinging 192.168.102.26 with 32 bytes of data:

Reply from 192.168.102.26: bytes=32 time<1ms TTL=126
Reply from 192.168.102.26: bytes=32 time<1ms TTL=126
Reply from 192.168.102.26: bytes=32 time<1ms TTL=126
Reply from 192.168.102.26: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.102.26:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.101.10

Pinging 192.168.101.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.101.10: bytes=32 time=1ms TTL=126
Reply from 192.168.101.10: bytes=32 time<1ms TTL=126
Reply from 192.168.101.10: bytes=32 time<1ms TTL=126

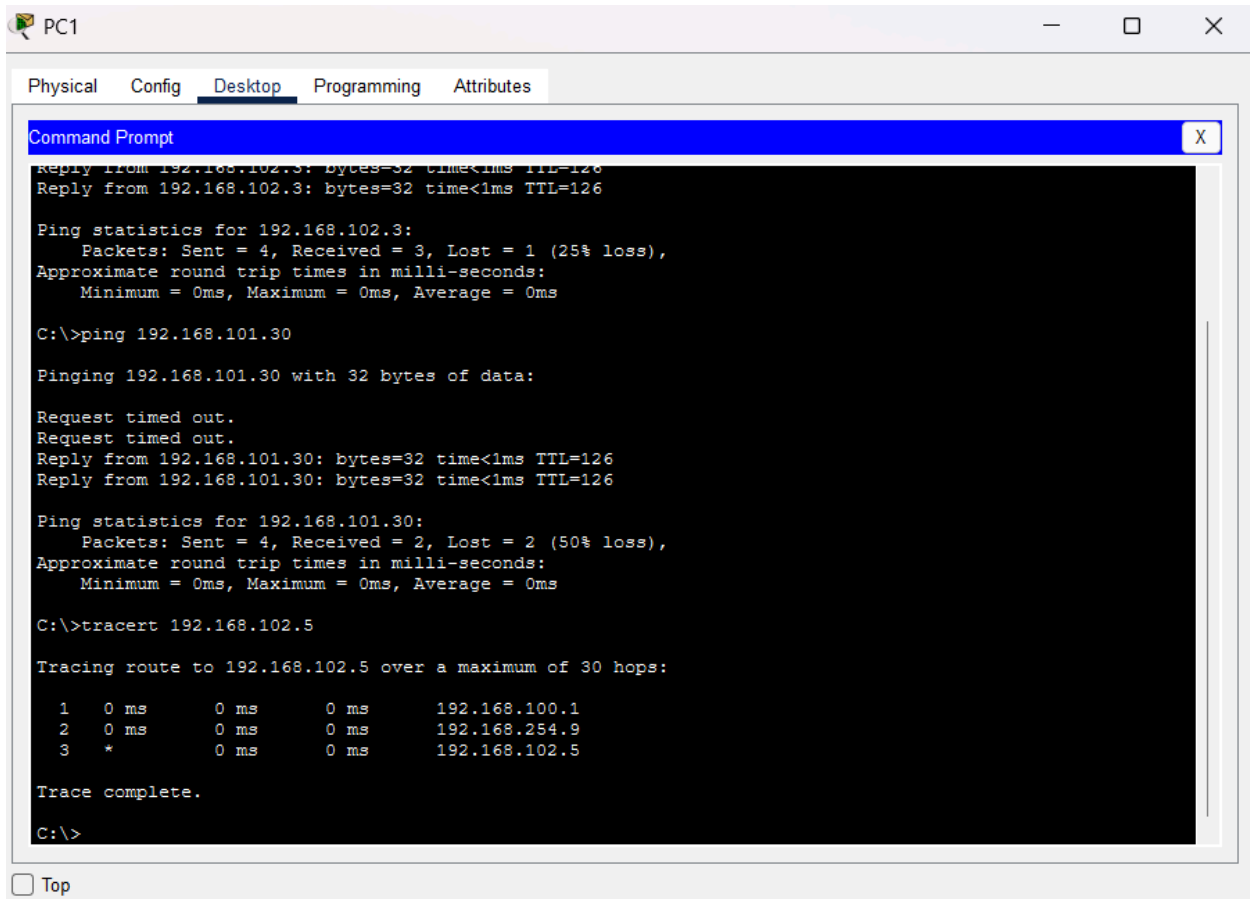
Ping statistics for 192.168.101.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

☐ Top

All around connectivity works. Feel free to pull my repo so you can test it yourself!

PC1 (Main Office - Tech Dept.) traceroute PC15 (New Office - Tech Dept.)



The screenshot shows a PC1 desktop environment with a window titled 'PC1'. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the results of several network tests:

```
Reply from 192.168.102.3: bytes=32 time<1ms TTL=126
Reply from 192.168.102.3: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.102.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.101.30

Pinging 192.168.101.30 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.30:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>tracert 192.168.102.5

Tracing route to 192.168.102.5 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.100.1
  2  0 ms    0 ms    0 ms    192.168.254.9
  3  *        0 ms    0 ms    192.168.102.5

Trace complete.

C:\>
```

At the bottom of the window, there is a 'Top' button.

The traceroute from PC1 in the Main Office Tech Department to PC15 in the New Office Tech Department demonstrates how OSPF is directing traffic between the two networks. The packet first reaches the Main Office gateway at 192.168.100.1, then is forwarded through the transit network to 192.168.254.9, before reaching the destination network at 192.168.102.5. This path corresponds with R1's routing table, where the O entry for 192.168.102.0/29 indicates that the New Office network was learned dynamically through OSPF via the next-hop address 192.168.254.9 on GigabitEthernet0/2. Together, the traceroute and routing table provide evidence that OSPF successfully learned the remote subnet and dynamically determined the path from the Main Office to the New Office.

Let's say we have OSPF, Default Routes, and Static Routes. Which will the router use?

When OSPF, static routes, and a default route are all configured on a router, the router does not simply choose one routing method over another. Instead, it uses the most specific route first, and when multiple routes to the same destination have the same prefix length, it compares their administrative distance. For example, a static route normally has an administrative distance of 1, while an OSPF route has 110, so if both provide the exact same destination and prefix, the router will prefer the static route. A default route (0.0.0.0/0) is used only when the routing table does not contain a more specific route for the destination. Therefore, in a network where OSPF has learned the specific route, that OSPF route will generally be used instead of the default route, while a manually configured static route to the same specific destination would take precedence over OSPF.

If that's the case, should we still configure multiple routes? Is there any benefit?

Yes, there can still be benefits to configuring multiple routes, especially for redundancy and different routing purposes. A dynamic protocol such as OSPF can handle the primary routing automatically, while static routes can be configured as backup paths using a higher administrative distance, allowing the router to switch to the backup route if the preferred OSPF path becomes unavailable. A default route can also serve as a gateway of last resort for destinations that are not specifically known in the routing table. Therefore, having multiple routing methods is not necessarily redundant or unnecessary; when designed properly, they can work together to provide reliability, backup connectivity, and controlled routing behavior.

Conclusion

The implementation of OSPF marked an important step in the development of the company's network as it expanded to include the new office. By removing the previously configured static and default routes and replacing them with single-area OSPF, the routers were able to dynamically discover and exchange information about the different networks within the topology. The routing tables are automatically populated with OSPF-learned routes, including the individual subnets of the Main Office and 2nd Office, demonstrating how OSPF can simplify route management as the network grows. Connectivity was also verified through ping and traceroute testing, confirming that traffic could successfully travel between the different office and server networks using dynamically learned routes. While static and default routes can still provide useful redundancy and specific routing control, OSPF provides a more scalable and maintainable approach for the company's expanding network.